Smart Library Request Workflow in ServiceNow

Technical Documentation — Identification Phase

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Academic & Technical Project Documentation

1. Introduction

Academic institutions rely on library services as a core component of their educational infrastructure. However, as student enrolment grows and book inventories expand, many institutions continue to operate library systems that depend heavily on manual processes — paper-based borrowing registers, verbal approvals, and ad hoc tracking methods. These approaches introduce operational inefficiencies that affect both library staff and end users.

The increasing availability of enterprise service management platforms offers a viable path to modernisation. ServiceNow, a cloud-based platform originally designed for IT service management, provides powerful tools for workflow automation, role-based access control, and data-driven reporting. Deploying a library management solution within ServiceNow enables academic institutions to digitise the entire borrowing lifecycle — from book discovery and request submission through to librarian approval and return tracking — within a single, governed platform.

This document constitutes the Identification Phase of the Smart Library Request Workflow project. It defines the problem context, articulates project objectives, identifies all relevant stakeholders, establishes the system scope, and enumerates the key functional requirements necessary to deliver a role-based Library Management System on ServiceNow.

2. Problem Identification

The following core deficiencies have been identified in conventional, manually operated library systems within academic settings.

2.1 Absence of Real-Time Book Availability Information

In manual systems, students and faculty members have no reliable mechanism for verifying whether a particular title is currently available on the shelf or is held by another borrower. The resulting uncertainty causes unnecessary visits to the library and contributes to borrower frustration. Without a centralised, live inventory record, staff must also invest significant time responding to availability enquiries.

2.2 Inefficient Approval and Borrowing Workflows

The absence of a structured approval process means that borrowing requests are typically handled informally — through verbal confirmation, handwritten forms, or email correspondence. These channels are prone to miscommunication, are difficult to audit, and cannot be easily escalated or reassigned when a librarian is unavailable. The lack of standardisation results in inconsistent service delivery across different staff members and shifts.

2.3 Inability to Track Borrow Requests and Return Obligations

Manual registers and spreadsheets provide limited visibility into the status of active borrow requests. It is difficult for librarians to identify overdue books, enforce return deadlines, or generate reliable summaries of borrowing patterns. The absence of automated alerts means that overdue items frequently go unnoticed until periodic physical audits are conducted, by which point the administrative burden of reconciliation has grown substantially.

2.4 Limited Reporting and Decision-Making Support

Without a structured data system, library administrators are unable to generate actionable reports on collection utilisation, peak demand periods, or student borrowing behaviour. This limits the institution's capacity to make informed procurement decisions, identify under-utilised resources, or justify budget allocations to management.

3. Project Objectives

The Smart Library Request Workflow project is guided by the following objectives, which collectively address the deficiencies identified in Section 2.

3.1 Implement a Role-Based Library Management System

The system will support two distinct user roles — Student and Librarian — each with clearly defined permissions and access rights. Students will be authorised to submit borrowing requests for available titles. Librarians will be authorised to review, approve, or reject those requests, manage the book inventory, and monitor the status of all active and historical borrow records.

3.2 Automate Approval Workflows

ServiceNow workflows will be configured to route newly submitted borrow requests automatically to the appropriate librarian queue. Automated notifications will be dispatched to both the requesting student and the assigned librarian at each stage of the approval process, eliminating the need for manual follow-up and reducing response latency.

3.3 Provide Real-Time Inventory Visibility

The Book table will maintain an up-to-date availability status for each title. When a borrow request is approved, the system will automatically mark the corresponding book as unavailable. Upon return confirmation, the status will revert to available. This real-time synchronisation ensures that students always have accurate information before submitting a request.

3.4 Enforce Access Control Through ACLs

Access Control Lists (ACLs) will be applied to both the Book table and the Borrow Request table, ensuring that students can only view and create their own records, while librarians retain full administrative visibility. This separation of privilege protects data integrity and prevents unauthorised modifications.

3.5 Enable Reporting and Analytics

The system will provide pre-configured reports — including a Most Borrowed Books report — to support library management decisions. These reports will draw on transactional data generated by the workflow, enabling trend analysis, demand forecasting, and resource planning.

4. Stakeholder Identification

The following stakeholders have been identified as directly or indirectly affected by the Smart Library Request Workflow. Understanding their roles and expectations is essential to ensuring that the system design meets institutional requirements and delivers measurable value.

Stakeholder	Primary Role	Key Expectations

Students	End users who search for, request, and return books through the self-service portal.	Intuitive interface, real-time availability status, and timely notifications on request outcomes.
Librarians	Operational administrators who manage the book inventory, review and process borrow requests, and oversee returns.	A centralised dashboard for managing requests, automated alerts for pending actions, and tools to enforce borrowing policies.
System Administrators	Platform custodians responsible for configuring roles, ACLs, workflows, and maintaining system integrity.	Stable, maintainable configuration; clear access controls; and adherence to institutional security policies.
Faculty / Management	Institutional decision-makers who rely on library data to inform procurement and resource planning strategies.	Accurate and timely reports on collection utilisation, student borrowing trends, and overall library performance.
IT Support Team	Technical support personnel who manage the ServiceNow environment, troubleshoot issues, and support system upgrades.	Well-documented configuration, minimal custom code, and a stable architecture that is easy to maintain and extend.

5. System Scope

The Smart Library Request Workflow encompasses the following functional domains. Each domain represents a discrete area of system capability that will be designed, configured, and tested within the ServiceNow platform.

5.1 Book Management

The system will maintain a centralised repository of all library holdings through a custom Book table. Each record will capture the book title, author, category, current availability status, and any additional metadata required for catalogue management. Librarians will be granted full create, read, update, and delete privileges on this table, while students will have read-only access limited to available titles.

5.2 Borrow Request Management

A dedicated Borrow Request table will record every borrowing transaction initiated within the system. Each record will capture the requesting student, the book being requested, the date of submission, the current approval status, and the expected return date. The lifecycle of a borrow request — from submission through approval or rejection to book return — will be governed entirely within this table.

5.3 Automated Workflow Approvals

ServiceNow Flow Designer will be used to automate the approval routing logic. When a student submits a borrow request, the workflow will automatically assign the task to the librarian queue, send an acknowledgement notification to the student, and generate a task notification for the librarian. Approval or rejection actions taken by the librarian will trigger corresponding notifications and status updates without any manual intervention.

5.4 Role-Based Access Control

Two custom roles will be defined within ServiceNow — lib_student and lib_librarian. Access Control Lists will be applied to both custom tables to enforce the principle of least privilege. Students will be restricted to submitting and viewing their own requests; librarians will have comprehensive access to all records. Administrative functions such as role assignment and ACL management will remain the exclusive domain of system administrators.

5.5 Reporting and Analytics

The system will include pre-built report definitions that enable library management to monitor operational performance. Core reports will include a Most Borrowed Books summary, active borrow request counts, and overdue request listings. These reports will be accessible to librarians and faculty through the standard ServiceNow reporting interface, with scheduling options available for periodic automated distribution.

6. Key Requirements

The following table enumerates the key functional and technical requirements that the system must satisfy to achieve the stated objectives. Each requirement is assigned a unique identifier for traceability throughout the development and testing phases.

Requirement ID	Requirement	Description
REQ-01	User Role Definition	Two roles must be created within ServiceNow: lib_student for students and lib_librarian for librarians. Each role governs all access permissions within the system.
REQ-02	Book Table	A custom table (u_library_book) must be created with fields for title, author, category, and availability status. This table serves as the master record for all library holdings.
REQ-03	Borrow Request Table	A custom table (u_borrow_request) must be created with fields for the

		requesting student, book reference, request date, approval status, and return date.
REQ-04	Workflow Automation	A ServiceNow flow must be configured to route borrow requests to the librarian queue, send automated notifications, and update record statuses upon approval or rejection.
REQ-05	Access Control Lists	ACLs must restrict student access to their own records only. Librarians must have full visibility and edit rights across all records in both tables.
REQ-06	Most Borrowed Books Report	A report must be configured to display the titles with the highest number of approved borrow requests, ranked in descending order by borrow frequency.
REQ-07	Real-Time Availability	The availability field on the Book record must update automatically when a borrow request is approved and revert to available when the return is confirmed by the librarian.
REQ-08	Notification Engine	Automated email or in-platform notifications must be sent to students upon request submission, approval, and rejection. Librarians must receive alerts when new requests require action.

7. Expected Outcomes

Upon successful implementation of the Smart Library Request Workflow, the institution can reasonably expect the following outcomes.

7.1 Improved Operational Efficiency

The elimination of manual, paper-based processes will reduce the time required to process a borrow request from initiation to approval. Automated workflow routing will ensure that requests are actioned promptly, reducing the risk of requests being overlooked or delayed due to staffing gaps.

7.2 Reduced Manual Effort for Library Staff

Librarians will no longer be required to manually log requests, maintain borrowing registers, or send follow-up reminders to students with overdue books. The system will perform these functions automatically, freeing library staff to focus on higher-value activities such as collection development and user support.

7.3 Transparent and Auditable Book Tracking

Every borrowing transaction will be recorded within the Borrow Request table, creating a complete and auditable history of each book's movement. Administrators and librarians will be able to determine the current holder of any title, review the full borrowing history of a student, and identify books that have been outstanding beyond their expected return date.

7.4 Enhanced Decision-Making Through Reporting

Institutional management will gain access to accurate, data-driven insights into library operations. Reports on the most borrowed titles, borrowing frequency by category, and request volume trends will inform procurement decisions, guide collection policy, and support the justification of library budget allocations to senior stakeholders.